Skills Summary

Equal Parts and Fractions

Use this reference while completing your worksheet or when studying.

Check First: Are the Parts Equal?

Rule: A fraction only names *equal* parts. Before writing any fraction, ask: **are all the parts the same size?** If one part is bigger than another, the parts do not have a fraction name yet.

Example: A paper strip is cut into parts that are 3 cm, 3 cm, and 6 cm long. Do the parts show thirds?
Step 1. To call each part one third, all three parts must be the same length — so compare the part lengths before naming any fraction.
Step 2. Two parts are 3 cm, but the last is 6 cm: one part is longer than the others.
The lengths are not all the same, $\mathbf{3 < 6}$, so the strip does *not* show thirds.

Try it: A ribbon is cut into parts that are 4 cm, 5 cm, and 4 cm long. Do the parts show thirds?

check: no — $\mathbf{4 < 5}$, the parts are not equal

(!) Watch out: Equal parts can have *different shapes*. Equal means the same amount of space (or length), not the same look.

Write the Fraction from an Equal-Parts Model

Rule: When the parts *are* equal: fraction shaded $= \dfrac{\text{shaded parts}}{\text{equal parts in all}}$

Example: A rectangle is cut into 8 equal parts, and 3 parts are shaded. What fraction is shaded?
Step 1. The parts are equal, so a fraction is allowed: count shaded parts for the top number and all the parts for the bottom number.
Step 2. Shaded parts: 3. Equal parts in all: 8.
So the shaded fraction is $\mathbf{\frac{3}{8}}$

Try it: A circle is cut into 6 equal parts, and 5 parts are shaded. What fraction is shaded?

check: $\mathbf{\frac{5}{6}}$

Fix an Unequal Model: Count Equal Squares

Rule: If the parts are *not* equal, do not count the parts! Cover the whole with same-size small squares, then count: true fraction $= \dfrac{\text{squares shaded}}{\text{squares in all}}$

Example: A rectangle made of 2 rows of 5 small squares is split by heavy lines into four parts that are *not* equal. The shaded part covers 2 small squares. What fraction is really shaded?

Step 1. The parts are not equal, so parts cannot be counted — count the same-size small squares instead.

Step 2. Squares in all: $2 \times 5 = 10$. Squares shaded: 2, so the fraction is $\frac{2}{10}$, which is the same as one fifth.

The true shaded fraction is $\mathbf{\frac{1}{5}}$

Try it: A rectangle made of 3 rows of 4 small squares is split into five unequal parts. The shaded part covers 3 small squares. What fraction is really shaded?

check: $\mathbf{\frac{1}{4}}$